

KATYAYANI UPADHYAY

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Profile Summary

AI/ML and Data Science Engineer building agentic GenAI systems and the data pipelines that keep them trustworthy. Shipped a live agentic RAG copilot that lifted **tool-call success from 0% to 100%** and citation coverage to **100%** over a retrieval-only baseline; operationalised human-in-the-loop validation for production catalogue at GobbleCube; cut day-ahead electricity load-forecast error **52%** vs baseline with a dbt + LightGBM pipeline featuring A/B testing (CUPED) and DiD causal analysis.

Education

A.P.J. Abdul Kalam Technical University

2026

B.Tech in Computer Science (Artificial Intelligence & Machine Learning specialized)

Lucknow, India

- CGPA: 8.56 | Relevant Coursework: Machine Learning, Deep Learning, NLP, DSA, Computer Vision

Work Experience

GobbleCube

June 2026 – Present

AI Engineer Intern

Gurugram, India

- Built the **first production-scale human validation layer** over an ML attribute extraction (mistral:7b) — a three-role FastAPI + React system on PostgreSQL spanning **100k SKUs**, processing **10k review decisions**.
- Recovered **~240K records (~27%)** lost to a join defect and drove schema drift to **0** (119 attributes resolved) via a required Pydantic validation gate and a 12-rule canonicalisation engine passing **161/161 tests**.

NIELIT (National Institute of Electronics & Information Technology)

August 2025 – February 2026

AI/ML Engineer Intern

Lucknow, India

- Achieved **80%+ multi-class emotion-classification accuracy** across **50,000+ audio samples** with a CNN + Librosa MFCC pipeline, deployed as a Flask API with **sub-200ms inference** and MLflow experiment tracking.
- Engineered an **autonomous multi-agent research assistant** (LangGraph, RAG, Groq, FastAPI, Docker) synthesising reports from **20+ real-time web sources**, cutting research-synthesis time by **70%**.

Projects

Category Unit-of-Measure Validation Pipeline | *Python, SQL, PostgreSQL, LLM, Regex* GobbleCube (Internal)

- Audited and corrected **33,447 machine-generated category-to-unit rules** via a **5-tier deterministic pipeline** grounded in Legal Metrology Rules 2011 — **322 categories human-reviewed, 65 disagreements adjudicated**.
- Caught and rejected a **circular-validation error** in a prior analysis, establishing an independent benchmark where the pipeline reproduced **26 of 40** human-verified corrections across disputed categories covering **15,937 products**.

QuickCommerce Copilot | *LangGraph, MCP, Qdrant, RAGAS, FastAPI, React, TypeScript, Docker*

[GitHub](#) | [Live](#)

- Engineered a **live agentic RAG assistant** pairing policy retrieval with **typed MCP tool calls** — tool-call success **0% → 100%**, citation coverage **50% → 100%** vs a retrieval-only baseline, RAGAS faithfulness held at **0.99**.
- Productionised it with a decide-act-reflect **LangGraph loop**, **citation guardrails** that refuse rather than guess, a **3-provider LLM failover chain**, and a **GitHub Actions eval gate** — shipped in Docker with a React/TypeScript agent-trace UI.

GridLoad Demand & Experiments | *Python, dbt, DuckDB, LightGBM, statsmodels, Streamlit*

[GitHub](#) | [Live](#)

- Cut day-ahead hourly load-forecast error by **52%** (WAPE **2.16%** vs **4.49%** seasonal-naive) over **15k hours** of real Spanish grid data, via LightGBM with a leakage-free rolling-origin backtest on a dbt + DuckDB warehouse (**56/56 tests**).
- Ran a simulated demand-response **A/B analysis** — power analysis (**MDE 4.2%**), **SRM check**, **CUPED (94.7%** variance reduction), ship/no-ship doc — plus a **DiD** lockdown study reporting a non-significant **-2.2%** effect.

Technical Skills

GenAI & Agentic AI: Agentic RAG, Multi-Agent Systems, LangGraph, MCP (Model Context Protocol), RAGAS (RAG/LLM Evaluation), Qdrant, LLM Orchestration, Prompt Engineering, Guardrails, HuggingFace Embeddings

Data Science & Statistics: Time-Series Forecasting, A/B Testing, Power Analysis & MDE, SRM, CUPED, Difference-in-Differences, Causal Inference, EDA, Feature Engineering

Data Engineering: SQL, PostgreSQL, dbt, DuckDB, ETL & Data Pipelines, Data-Quality Testing, Schema Design, Pydantic

AI / ML: Deep Learning, CNNs, Transfer Learning, NLP, LightGBM, MLflow, Hyperparameter Tuning

MLOps & Deployment: FastAPI, Docker, GitHub Actions (CI/CD), Automated Eval Gates, Streamlit, Render, AWS (EC2, RDS)

Programming & Libraries: Python, SQL, Java, TypeScript, React, PyTorch, TensorFlow, Scikit-learn, pandas, NumPy, statsmodels, Librosa

Languages: English (Advanced), Hindi (Native), Spanish (Intermediate)